TÜVNORD

EC-Type Examination Certificate (Module B)

Certificate No.: M26016-B-V01
Expiry date: 2031-04-12, Date of (re-)issue: 2026-04-13

This is to certify that the TÜV NORD Systems GmbH & Co. KG – Certification Body (NB0045) did undertake the relevant type approval procedures for the type of equipment identified below which was found to be in compliance with the requirements of
Marine Equipment Directive (MED) 2014/90/EU
subject to any conditions in the schedule attached hereto.

Product Identification:	**LED.11IGN.A60**
Product Type:	Firedoors
Product Description:	Single leaf sliding firedoor (Class A-60)
Manufacturer:	**Lethe Exterior Doors GmbH** Marschgehren 14 28779 Bremen Germany
Authorized Representative:	-/-

Regulation Item:	**MED/3.16 (Row 1 of 1)**
USCG Approval Category:	164.136/EC0045
Specified Standards:	SOLAS 74 Reg. II-2/9 IMO MSC.1/Circ.1511 IMO MSC.1/Circ.1319 IMO Res. MSC.307(88)-(2010 FTP Code), as amended
Directive Reference:	Marine Equipment Directive (MED) 2014/90/EU, Regulation 2025/1533/EU

The attached annex (schedule of appraisal) forms part of this certificate.
This certificate remains valid unless suspended, expired or withdrawn, provided the conditions in the attached schedule are complied with.

Digital unterschrieben von Capraro Sebastian
Datum: 2026.04.13
15:01:30 +02'00'

Sebastian Capraro
Deputy head of certification program MED

Note 1: This certificate will not be valid if the manufacturer makes any changes or modifications to the approved type of equipment, which have not been notified to, and agreed with the notified body named on this certificate.
Note 2: During the period of validity of this certificate the applicable regulations (international conventions, the relevant resolutions and circulars of the IMO) and testing standards of the Commission Implementing Regulation may change, therefore the product conformity may need to be re-assessed by the notified body.
Note 3: The Mark of Conformity may only be affixed to the above type approved equipment and a Manufacturer's Declaration of Conformity issued when the production-control phase module (D, E, or F) of the Directive is fully complied with.
Note 4: In case limitations of use apply, these should be indicated of in the Annex.
Note 5: An U.S. Coast Guard approval number will be assigned to the equipment when the production module has been completed and will appear on the production module certificate (module D, E or F), as allowed by the “Agreement between the European Community and the United States of America on the Mutual Recognition of Certificates of Conformity for Marine Equipment”

XXXX/YYYY

Schedule of appraisal

Alternative (local) Trade Names:

none

Approval documentation

- Test report DMT-DO-53-206-R2 issued by DMT GmbH & Co. KG, Prüfstelle für Brandschutz, Germany on 2025-05-06 (Revision 2026-03-26)
- Test report DMT-DO-53-229-R1 issued by DMT GmbH & Co. KG, Prüfstelle für Brandschutz, Germany on 2026-03-05 (Revision 2026-03-26)
- Assessment 8123901305 „LED.11IGN.A60“ issued by the DMT GmbH & Co. KG, Test Laboratory for Fire Protection on 2026-03-12
- Drawing “KD-0226916-REV01.4 DMT-DO-53-206” issued Lethe Exterior Doors GmbH on 2025-09-17, Revision 1.4

Conducted tests:

- Tested according to IMO 2010 FTP Code Annex 1: Part 3

Technical principles:

The product “LED.11IGN.A60” is a single-leaf sliding door that is classified as A-60 and is designed for a clear opening of 1249 mm x 2418 mm (W x H). Its main components include the door leaf, the frame, the slide track and the sealing material.

The door leaf, measuring 1310 mm x 2397 mm (W x H), consists of four layers of "SeaPan WP 15/15", which are bonded with the adhesive "LR Cerammatrix 01-50". The total thickness of the door leaf is 60 mm. For reinforcement, the door leaf is framed by a square tube measuring 30 mm x 30 mm with a material thickness of 2 mm. The square tubes are welded. Strips of “Thermax SN” are fitted (via self-cutting screws) on both sides of the door leaf along its entire height at the side of the inlet. The edge of the door leaf on the side of the inlet is covered with steel angles (45 mm x 32 mm, material thickness 0.75 mm). The angles are fastened via self-cutting screws.

The frame of the sliding door consists of the inlet, the counter-inlet, the threshold, and the door housing.

The reveal of the inlet is lined with two strips of “Thermax SN” along its entire height. These strips are attached using steel angles (56 mm x 34 mm, 2 mm thick) fastened with self-tapping screws reinforced by a square tubes (60 mm x 25 mm, 2 mm material thickness). The square tubes are clad with two layers of “SeaPan WP 50/60.” The individual layers are bonded together using “LR Cerammatrix 01-50” adhesive and screwed via self-tapping screws to the square tube.

The sliding door housing, with dimensions 1511 mm x 2768 mm (W x H), accommodates the door panel “when open”. It consist of two layers of “SeaPan WP 50/60”, which are bonded together with “LR Cerammatrix 01-50” and connected to the square tubes using self-tapping screws. In the center is a vertical reinforcement consisting of a square tube (60 mm x 25 mm, material thickness 2 mm). The back of the housing is reinforced with a square tube (80 mm x 40 mm, material thickness 2 mm).

The reveal of the counter-inlet is lined with two strips of “Thermax SN” along its entire height. The counter-inlet is reinforced by a square tube measuring 120 mm x 60 mm (material thickness 2 mm), which is clad with two layers of “SeaPan WP 50/60”. The individual layers are bonded together with “LR Cerammatrix 01-50” adhesive and screwed to the square tube.

The threshold is made of U-shaped steel measuring 80 x 200 x 80 mm (material thickness 6 mm) and is secured to the bulkhead with ten M12 screws (via welded studs). Below the threshold is an arrangement of two mounting brackets that can be aligned against the bulkhead using adjusting screws. Five adjustable feet (maximum distance of 690 mm) are attached to the mounting brackets to support the threshold.

The slide track was implemented using the “HST-S” product from Heise GmbH. The main components of the system are made of aluminum; the slide rail is made of steel, and the guide roller is made of steel and plastic. The slide track is secured via an L-profile to a square profile, which is attached to the weld-on bolts of the bulkhead using an additional steel bracket. The connection to the door leaf was implemented using two hangers. The slide track housing is clad with two layers of “SeaPan WP 50/60” on both sides and the top. A single layer of “SeaPan WP 15/15” is applied to the bottom of the housing.

"Promseal PL" is used as the sealing material. It is applied symmetrically to the inside of the guide rail covers and symmetrically on both sides of the frame openings (inlet and counter-inlet). In addition, the sealing material is fitted to the door leaf. It is positioned on the side of the inlet, attached to the “Thermax SN” strips, and on the top of the door leaf, to the right and left of the hanging mechanism.

Optionally, the fire door can be manufactured with a hose port. The hose port is located in the lower section of the door leaf, extends across the entire width of the door leaf, and has a maximum height of 135 mm. It consists of four layers of “SeaPan WP 15/15” bonded with “LR Cerammatrix 01-50” adhesive. The total thickness of the hose connection is 60 mm. For reinforcement, the hose connection is framed by a square tube measuring 30 mm x 30 mm with a wall thickness of 2 mm. The square tubes are welded. The connection between the hose connection and the door leaf is made using a linear guide (SHS20V1QZSS in combination with SHS20-5000L, manufactured by THL Co., Ltd.) and secured by a flat steel.

Applied materials:

Components of the insulating material

- SeaPan WP 15/15 - CBG Composites GmbH
  - CBG LtBC - CBG Composites GmbH
  - CBG LtGC - CBG Composites GmbH
  - LifeRock MW-180 - CBG Composites GmbH
- SeaPan WP 50/60 - CBG Composites GmbH
- THERMAX SN - Mineralka Austria GmbH

Sealing and Adhesive:

- Promaseal PL- Promat Research and Technology Centre
- LR Cerammatrix 01-50 - CBG Composites GmbH

Installation:

Assembly procedure, usage and maintenance on board as per manufacturer´s instruction, which has to be supplied together with the product.

Remarks:

The requirements of sections 4.1.1.4 until 4.1.1.9 of SOLAS 74 Reg. II-2/9 are not evaluated and analyzed within the scope of the certification.

The requirements set out in IMO MSC.1/Circ.1319, Part 2, “Doors with slightly larger dimensions”, are fulfilled.

Limitations / Acceptance on use of the product:

See under “Technical principles”

Comment to USCG Approval:

Limited to fire doors without windows or with total window area no more than 645 cm² in each door leaf, unless hose stream, heat flux, and/or temperature rise tests are conducted according to USCG approval guidance.

Approval limited to maximum door size tested. Doors must be used with a fire tested frame design.

Marking:

The product shall be permanently marked in accordance with Article 10 of the Council Directive 2014/90/EC of 23rd of July 2014 on marine equipment, e.g. certificate (approval) number, fire rating, etc.

The Mark of Conformity may only be affixed to the above type approved equipment and a Manufacturer's Declaration of Conformity issued when the production-control phase module (D, E, or F) of the Directive is fully complied with and controlled by a written inspection agreement with a notified body.